Istio Traffic Management Lab Report

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1. Introduction and Objectives

This lab exercise provides hands-on experience with Istio's traffic management capabilities in a Kubernetes environment. The primary goal is to understand how service mesh features can manage microservice deployments without modifying application code. The four topics covered are Request Routing, Traffic Shifting, Fault Injection, and Circuit Breaking. These capabilities are essential for production-grade microservice architectures where gradual rollouts, user-specific routing, resilience testing, and overload protection are critical requirements.

2. Environment Setup

The experiments were conducted on a Windows machine using Docker Desktop with its built-in Kubernetes cluster. Istio version 1.29.2 was installed with the default profile, which includes the control plane (istiod), ingress gateway, and observability addons such as Grafana, Kiali, Prometheus, Jaeger, and Zipkin. The Bookinfo sample application was pre-deployed in the default namespace with automatic sidecar injection enabled, resulting in all application pods showing 2 out of 2 containers ready.

The httpbin and fortio applications were used exclusively for the Circuit Breaking experiment. Access to the Bookinfo application was established through port-forwarding the bookinfo-gateway-istio service to localhost port 8080, as the environment uses the Kubernetes Gateway API deployment pattern rather than the traditional Istio Ingress Gateway direct access model.

3. Experiment One: Request Routing

3.1 Overview
The Request Routing experiment demonstrates how Istio can dynamically route HTTP requests to different versions of a microservice based on predefined rules. This is the foundation of all advanced traffic management features in Istio.

3.2 Execution Steps
First, the DestinationRule resources were applied to define subsets for each Bookinfo service version. The command used was kubectl apply with the GitHub raw URL for destination-rule-all-mtls.yaml, because the local samples directory was not present in the current working path. Then the virtual-service-all-v1.yaml was applied to route 100 percent of traffic to

version 1 of all services including productpage, details, reviews, and ratings.

After waiting a few seconds for configuration propagation, the Bookinfo product page was accessed at localhost:8080/productpage. Multiple refreshes confirmed that the reviews section consistently showed no star ratings, because reviews v1 does not call the ratings service.

Next, the virtual-service-reviews-test-v2.yaml was applied. This rule added a header match condition: if the end-user header equals jason, traffic is routed to reviews v2. Otherwise, it falls back to reviews v1. The user logged in as jason and refreshed the page. Black star ratings appeared next to each review. After signing out and logging in as a different user, the stars disappeared and the page returned to the v1 behavior.

3.3 Key Observations

The VirtualService and DestinationRule work together to achieve layer 7 routing. The DestinationRule defines the subsets based on pod labels, while the VirtualService defines the routing logic. This separation of concerns allows operators to change routing rules without touching the underlying deployment or service definitions.

4. Experiment Two: Traffic Shifting

4.1 Overview

Traffic Shifting demonstrates a canary deployment strategy where traffic is gradually migrated from an old service version to a new one using weight-based routing. This is fundamentally different from scaling pods, because the traffic distribution is controlled at the proxy level regardless of the actual pod replica counts.

4.2 Execution Steps

Starting from the all-v1 state, the virtual-service-reviews-50-v3.yaml was applied. This configuration set the weight to 50 for reviews v1 and 50 for reviews v3. The kubectl get virtualservice reviews command confirmed the weight values in the YAML output.

The product page was refreshed approximately ten times. Roughly half of the requests showed no stars (v1 behavior) and the other half showed red stars (v3 behavior). This confirmed that the Envoy sidecar proxies were distributing traffic according to the specified weights.

Finally, the virtual-service-reviews-v3.yaml was applied to send 100 percent of traffic to reviews v3. After this change, every refresh consistently displayed red star ratings. The reviews v1 and v3 pods remained running independently, proving that Istio routing is decoupled from pod scaling.

4.3 Key Observations

Weight-based routing enables safe production rollouts. Operators can monitor error rates and latency during the 50-50 phase, and if metrics are healthy, proceed to full migration. If problems arise, a quick rollback to the previous VirtualService restores the old behavior instantly.

5. Experiment Three: Fault Injection

5.1 Overview

Fault Injection allows operators to introduce artificial delays and HTTP errors into the request path to test application resilience. This experiment revealed a realistic microservice bug related to inconsistent timeout configurations across service boundaries.

5.2 Execution Steps

The experiment required the request routing setup from Experiment One as a prerequisite, meaning jason should route to reviews v2 while all other users route to reviews v1. The virtual-service-ratings-test-delay.yaml was applied to inject a 7-second fixed delay into the ratings service, but only for requests from user jason.

When logged in as jason and refreshing the product page, the reviews section displayed an error message stating that product reviews were currently unavailable. Using the browser's Developer Tools Network tab, the productpage request took approximately 6 seconds to complete, not 7 seconds.

The analysis revealed a hardcoded timeout mismatch. The reviews v2 service has a 10-second timeout when calling ratings, so the 7-second delay does not break that connection. However, the productpage service has a hardcoded 3-second timeout plus 1 retry for a total of 6 seconds when calling reviews. Therefore, productpage gives up before the reviews-to-ratings call completes. This is a classic cross-team microservice bug where upstream and downstream timeout assumptions are inconsistent.

To verify the fix, all traffic was shifted to reviews v3 using virtual-service-reviews-v3.yaml. The reviews v3 implementation reduces the ratings timeout from 10 seconds to 2.5 seconds. The delay rule was then edited from 7 seconds to 2 seconds using kubectl edit virtualservice ratings. After this change, jason's requests completed successfully with red stars displayed, because 2 seconds is less than the 2.5-second timeout.

Subsequently, the delay rule was removed and replaced with the virtual-service-ratings-test-abort.yaml, which injects an HTTP 500 abort for jason. The page loaded immediately without delay and showed the message "Ratings service is currently unavailable." Anonymous users or other logged-in users saw no error because they were routed to reviews v1, which does not call the ratings service at all.

5.3 Key Observations

Fault injection is a powerful non-intrusive testing tool. It exposes integration bugs that unit tests and integration tests in isolated environments often miss. The experiment also demonstrated that fixing a bug in v3 and using traffic shifting to migrate users is a safe deployment strategy.

6. Experiment Four: Circuit Breaking

6.1 Overview

Circuit Breaking protects backend services from being overwhelmed by excessive concurrent connections and requests. By setting connection pool limits and outlier detection parameters, Envoy can reject surplus traffic with HTTP 503 before it reaches the application container.

6.2 Execution Steps

The httpbin application was deployed as the backend service. A DestinationRule named httpbin was created with strict connection pool limits: maxConnections of 1 for TCP, http1MaxPendingRequests of 1, and maxRequestsPerConnection of 1. The outlierDetection section was configured with consecutive5xxErrors set to 1, interval of 1 second, baseEjectionTime of 3 minutes, and maxEjectionPercent of 100.

The fortio load testing client was deployed and a single curl request was sent to verify basic connectivity. The request returned HTTP 200 with the server header showing envoy, confirming the sidecar proxy was intercepting traffic.

Then a load test was executed with 2 concurrent connections and 20 total requests. The results showed 17 requests succeeded with HTTP 200 and 3 requests received HTTP 503. This indicated that the circuit breaker was starting to engage but still allowed most traffic through due to Envoy's internal tolerance.

When the concurrency was increased to 3 connections with 30 total requests, the results changed dramatically. Only 11 requests succeeded (36.7 percent) while 19 requests were rejected with 503 (63.3 percent). This confirmed that the circuit breaker was actively protecting the httpbin service from overload.

Finally, the Envoy statistics were queried from the fortio pod's istio-proxy container. The command pilot-agent request GET stats was used with grep filters for httpbin and pending. The output showed upstream_rq_pending_overflow with a value of 21, meaning 21 requests were flagged for circuit breaking.

6.3 Key Observations

Circuit breaking is essential for graceful degradation under load. The sharp increase in 503 responses from 15 percent at 2 concurrent connections to 63.3 percent at 3 concurrent connections demonstrates how quickly Envoy enforces limits once thresholds are crossed. In production, operators should avoid setting maxEjectionPercent to 100 percent to prevent complete pool evacuation, and should tune consecutive5xxErrors to 3 or higher to avoid ejecting healthy endpoints due to transient errors.

7. Challenges and Reflections

Several challenges were encountered during the lab. First, Docker Desktop initially failed to start because CPU virtualization was not enabled in BIOS. This required entering the BIOS settings and enabling Intel VT-x or AMD-V, then enabling the Virtual Machine Platform and Windows

Subsystem for Linux features in Windows.

Second, the istio-ingressgateway service was missing initially because the Istio installation profile did not include ingress gateway components. This was resolved by reinstalling Istio with the default profile and explicitly enabling the ingress gateway component.

Third, the environment uses the Kubernetes Gateway API for Bookinfo, meaning the traditional istio-ingressgateway port-forwarding approach did not work. The correct service to port-forward was bookinfo-gateway-istio, not istio-ingressgateway. This highlights the importance of understanding which gateway resource actually fronts the application in a given cluster.

Fourth, numerous PowerShell syntax differences caused command failures. The export command does not exist in PowerShell; environment variables must be set with $env: syntax. The heredoc syntax with EOF is not supported in PowerShell, so YAML content had to be written to files using Out-File or Here-String piping. Additionally, JSONPath expressions containing double quotes required backslash escaping in PowerShell, such as name==\"http2\".

Fifth, all local sample file paths failed because the working directory was C:\WINDOWS\system32 rather than the Istio installation root. The consistent solution was to use GitHub raw URLs for all sample YAML files.

These challenges reflect real-world platform differences. Linux-centric documentation often assumes Bash environments, but Windows developers must adapt commands for PowerShell behavior. Understanding these translation layers is a valuable DevOps skill.

8. Overall Summary

This lab successfully demonstrated four core Istio traffic management capabilities using the Bookinfo and httpbin sample applications. Request Routing showed how header-based rules can personalize user experiences. Traffic Shifting illustrated safe canary deployments through weight-based migration. Fault Injection uncovered a realistic timeout inconsistency bug without impacting production users. Circuit Breaking proved that connection pool limits can prevent cascade failures under concurrent load.

The key architectural insight is that Istio separates traffic management from application logic. All routing, shifting, fault injection, and circuit breaking policies are configured through Kubernetes custom resources (VirtualService, DestinationRule) and enforced by the Envoy sidecar proxies. Application developers do not need to implement these concerns in their code, and operators can change behavior dynamically without redeploying containers.

For future work, these techniques can be extended to include request timeouts, retries, mirroring, and mutual TLS policies. The observability tools installed alongside Istio (Kiali, Grafana, Jaeger) provide the necessary visibility to make informed traffic management decisions in production environments.

Screenshot:

```
PS C:\WINDOWS\system32> istioctl version
client version: 1.29.2
control plane version: 1.29.2
data plane version: 1.29.2 (7 proxies)
PS C:\WINDOWS\system32> kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:53335
CoreDNS is running at https://127.0.0.1:53335/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
PS C:\WINDOWS\system32> kubectl get pods
NAME                                     READY   STATUS    RESTARTS        AGE
bookinfo-gateway-istio-b767fd6f4-v5wsc   1/1     Running   2 (4m32s ago)   13d
details-v1-77d6bd5675-rkrcr              2/2     Running   1 (5m17s ago)   13d
productpage-v1-bb87ff47b-rdsv6           2/2     Running   1 (5m17s ago)   13d
ratings-v1-8589f64b4c-7g88m              2/2     Running   1 (5m17s ago)   13d
reviews-v1-8cf7b9cc5-cxz6g               2/2     Running   1 (5m17s ago)   13d
reviews-v2-67d565655f-hmxhc              2/2     Running   1 (5m17s ago)   13d
reviews-v3-d587fc9d7-tkdsm               2/2     Running   1 (5m17s ago)   13d
```

```
PS C:\WINDOWS\system32> kubectl delete destinationrule httpbin
destinationrule.networking.istio.io "httpbin" deleted from default namespace
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/httpbin/sam
ple-client/fortio-deploy.yaml
service "fortio" deleted from default namespace
deployment.apps "fortio-deploy" deleted from default namespace
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/httpbin/htt
pbin.yaml
serviceaccount "httpbin" deleted from default namespace
service "httpbin" deleted from default namespace
deployment.apps "httpbin" deleted from default namespace
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl exec "$env:FORTIO_POD" -c istio-proxy -- pilot-agent request GET stats 2>$null | findstr
 httpbin | findstr pending
cluster.outbound|8000||httpbin.default.svc.cluster.local;.circuit_breakers.default.rq_pending_open: 0
cluster.outbound|8000||httpbin.default.svc.cluster.local;.circuit_breakers.high.rq_pending_open: 0
cluster.outbound|8000||httpbin.default.svc.cluster.local;.upstream_rq_pending_active: 0
cluster.outbound|8000||httpbin.default.svc.cluster.local;.upstream_rq_pending_failure_eject: 0
cluster.outbound|8000||httpbin.default.svc.cluster.local;.upstream_rq_pending_overflow: 19
cluster.outbound|8000||httpbin.default.svc.cluster.local;.upstream_rq_pending_total: 22
PS C:\WINDOWS\system32>
```

```
{"ts":1779526228.742994,"level":"warn","r":51,"file":"http_client.go","line":1151,"msg"
:"Non ok http code","code":503,"status":"HTTP/1.1 503","thread":0,"run":0}
{"ts":1779526228.743783,"level":"warn","r":51,"file":"http_client.go","line":1151,"msg"
:"Non ok http code","code":503,"status":"HTTP/1.1 503","thread":0,"run":0}
Ended after 22.153221ms : 30 calls. qps=1354.2
Aggregated Function Time : count 30 avg 0.0021308693 +/- 0.001896 min 0.000466888 max 0
.008237961 sum 0.06392608
# range, mid point, percentile, count
>= 0.000466888 <= 0.001 , 0.000733444 , 40.00, 12
> 0.001 <= 0.002 , 0.0015 , 70.00, 9
> 0.002 <= 0.003 , 0.0025 , 73.33, 1
> 0.003 <= 0.004 , 0.0035 , 83.33, 3
> 0.004 <= 0.005 , 0.0045 , 93.33, 3
> 0.006 <= 0.007 , 0.0065 , 96.67, 1
> 0.008 <= 0.00823796 , 0.00811898 , 100.00, 1
# target 50% 0.00133333
# target 75% 0.00316667
# target 90% 0.00466667
# target 99% 0.00816657
# target 99.9% 0.00823082
Error cases : count 23 avg 0.0014479115 +/- 0.001538 min 0.000466888 max 0.008237961 su
m 0.033301964
# range, mid point, percentile, count
>= 0.000466888 <= 0.001 , 0.000733444 , 52.17, 12
> 0.001 <= 0.002 , 0.0015 , 91.30, 9
> 0.002 <= 0.003 , 0.0025 , 95.65, 1
> 0.008 <= 0.00823796 , 0.00811898 , 100.00, 1
# target 50% 0.000975768
# target 75% 0.00158333
# target 90% 0.00196667
# target 99% 0.00818323
# target 99.9% 0.00823249
# Socket and IP used for each connection:
[0]   8 socket used, resolved to 10.96.95.119:8000, connection timing : count 8 avg 0.0
00157854 +/- 6.91e-05 min 8.2741e-05 max 0.000253957 sum 0.001262832
[1]   8 socket used, resolved to 10.96.95.119:8000, connection timing : count 8 avg 0.0
001745095 +/- 0.0001045 min 8.2665e-05 max 0.0003637 sum 0.001396076
[2]   7 socket used, resolved to 10.96.95.119:8000, connection timing : count 7 avg 0.0
0016240171 +/- 8.306e-05 min 3.7615e-05 max 0.000292958 sum 0.001136812
Connection time (s) : count 23 avg 0.0001650313 +/- 8.723e-05 min 3.7615e-05 max 0.0003
637 sum 0.00379572
Sockets used: 23 (for perfect keepalive, would be 3)
Uniform: false, Jitter: false, Catchup allowed: true
IP addresses distribution:
10.96.95.119:8000: 23
Code 200 : 7 (23.3 %)
Code 503 : 23 (76.7 %)
Response Header Sizes : count 30 avg 57.166667 +/- 103.6 min 0 max 245 sum 1715
Response Body/Total Sizes : count 30 avg 358.63333 +/- 282.7 min 153 max 866 sum 10759
All done 30 calls (plus 0 warmup) 2.131 ms avg, 1354.2 qps
PS C:\WINDOWS\system32>
```

```
>= 0.000382995 <= 0.001 , 0.000691498 , 20.00, 4
> 0.001 <= 0.002 , 0.0015 , 35.00, 3
> 0.002 <= 0.003 , 0.0025 , 55.00, 4
> 0.003 <= 0.004 , 0.0035 , 80.00, 5
> 0.004 <= 0.005 , 0.0045 , 85.00, 1
> 0.005 <= 0.006 , 0.0055 , 95.00, 2
> 0.007 <= 0.00785399 , 0.00742699 , 100.00, 1
# target 50% 0.00275
# target 75% 0.0038
# target 90% 0.0055
# target 99% 0.00768319
# target 99.9% 0.00783691
Error cases : count 7 avg 0.0012048974 +/- 0.0007173 min 0.000382995 max 0.002202145 su
m 0.008434282
# range, mid point, percentile, count
>= 0.000382995 <= 0.001 , 0.000691498 , 57.14, 4
> 0.001 <= 0.002 , 0.0015 , 71.43, 1
> 0.002 <= 0.00220214 , 0.00210107 , 100.00, 2
# target 50% 0.000897166
# target 75% 0.00202527
# target 90% 0.00213139
# target 99% 0.00219507
# target 99.9% 0.00220144
# Socket and IP used for each connection:
[0]   6 socket used, resolved to 10.96.95.119:8000, connection timing : count 6 avg 0.0
0022065917 +/- 3.545e-05 min 0.000174114 max 0.000268767 sum 0.001323955
[1]   3 socket used, resolved to 10.96.95.119:8000, connection timing : count 3 avg 0.0
00198029 +/- 4.078e-05 min 0.000143706 max 0.000241973 sum 0.000594087
Connection time (s) : count 9 avg 0.00021311578 +/- 3.881e-05 min 0.000143706 max 0.000
268767 sum 0.001918042
Sockets used: 9 (for perfect keepalive, would be 2)
Uniform: false, Jitter: false, Catchup allowed: true
IP addresses distribution:
10.96.95.119:8000: 9
Code 200 : 13 (65.0 %)
Code 503 : 7 (35.0 %)
Response Header Sizes : count 20 avg 159.25 +/- 116.9 min 0 max 245 sum 3185
Response Body/Total Sizes : count 20 avg 647.25 +/- 298.1 min 241 max 866 sum 12945
All done 20 calls (plus 0 warmup) 2.872 ms avg, 672.9 qps
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl exec "$env:FORTIO_POD" -c fortio -- /usr/bin/fortio loa
d -c 2 -qps 0 -n 20 -loglevel Warning http://httpbin:8000/get
{"ts":1779526172.596379,"level":"info","r":1,"file":"logger.go","line":298,"msg":"Log l
evel is now 3 Warning (was 2 Info)"}
Fortio 1.69.5 running at 0 queries per second, 16->16 procs, for 20 calls: http://httpb
in:8000/get
Starting at max qps with 2 thread(s) [gomax 16] for exactly 20 calls (10 per thread + 0
)
{"ts":1779526172.609987,"level":"warn","r":67,"file":"http_client.go","line":1151,"msg"
:"Non ok http code","code":503,"status":"HTTP/1.1 503","thread":0,"run":0}
{"ts":1779526172.612988,"level":"warn","r":68,"file":"http_client.go","line":1151,"msg"
:"Non ok http code","code":503,"status":"HTTP/1.1 503","thread":1,"run":0}
{"ts":1779526172.615479,"level":"warn","r":67,"file":"http_client.go","line":1151,"msg"
:"Non ok http code","code":503,"status":"HTTP/1.1 503","thread":0,"run":0}
{"ts":1779526172.619018,"level":"warn","r":67,"file":"http_client.go","line":1151,"msg"
:"Non ok http code","code":503,"status":"HTTP/1.1 503","thread":0,"run":0}
{"ts":1779526172.621018,"level":"warn","r":67,"file":"http_client.go","line":1151,"msg"
:"Non ok http code","code":503,"status":"HTTP/1.1 503","thread":0,"run":0}
{"ts":1779526172.623718,"level":"warn","r":68,"file":"http_client.go","line":1151,"msg"
:"Non ok http code","code":503,"status":"HTTP/1.1 503","thread":1,"run":0}
{"ts":1779526172.625836,"level":"warn","r":67,"file":"http_client.go","line":1151,"msg"
:"Non ok http code","code":503,"status":"HTTP/1.1 503","thread":0,"run":0}
Ended after 29.720334ms : 20 calls. qps=672.94
Aggregated Function Time : count 20 avg 0.0028720161 +/- 0.001804 min 0.000382995 max 0
.007853986 sum 0.057440322
# range, mid point, percentile, count
>= 0.000382995 <= 0.001 , 0.000691498 , 20.00, 4
> 0.001 <= 0.002 , 0.0015 , 35.00, 3
> 0.002 <= 0.003 , 0.0025 , 55.00, 4
> 0.003 <= 0.004 , 0.0035 , 80.00, 5
> 0.004 <= 0.005 , 0.0045 , 85.00, 1
> 0.005 <= 0.006 , 0.0055 , 95.00, 2
> 0.007 <= 0.00785399 , 0.00742699 , 100.00, 1
# target 50% 0.00275
# target 75% 0.0038
# target 90% 0.0055
# target 99% 0.00768319
# target 99.9% 0.00783691
Error cases : count 7 avg 0.0012048974 +/- 0.0007173 min 0.000382995 max 0.002202145 su
m 0.008434282
# range, mid point, percentile, count
>= 0.000382995 <= 0.001 , 0.000691498 , 57.14, 4
> 0.001 <= 0.002 , 0.0015 , 71.43, 1
> 0.002 <= 0.00220214 , 0.00210107 , 100.00, 2
# target 50% 0.000897166
# target 75% 0.00202527
# target 90% 0.00213139
# target 99% 0.00219507
# target 99.9% 0.00220144
```

```
PS C:\WINDOWS\system32> kubectl exec "$env:FORTIO_POD" -c fortio -- /usr/bin/fortio curl -quiet http://httpbin:8000/get
HTTP/1.1 200 OK
access-control-allow-credentials: true
access-control-allow-origin: *
content-type: application/json; charset=utf-8
date: Sat, 23 May 2026 08:48:42 GMT
content-length: 621
x-envoy-upstream-service-time: 10
server: envoy

{
  "args": {},
  "headers": {
    "Host": [
      "httpbin:8000"
    ],
    "User-Agent": [
      "fortio.org/fortio-1.69.5"
    ],
    "X-Envoy-Attempt-Count": [
      "1"
    ],
    "X-Forwarded-Client-Cert": [
      "By=spiffe://cluster.local/ns/default/sa/httpbin;Hash=b2930cd745891448a0a21c4445aed5711c66e0f10a46a6ba71ffb69a8a1d
07de;Subject=\"\";URI=spiffe://cluster.local/ns/default/sa/default"
    ],
    "X-Forwarded-Proto": [
      "http"
    ],
    "X-Request-Id": [
      "a262458e-fb16-4166-9800-6b38b42e708f"
    ]
  },
  "method": "GET",
  "origin": "127.0.0.6:38845",
  "url": "http://httpbin:8000/get"
}
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/httpbin/samp
le-client/fortio-deploy.yaml
service/fortio created
deployment.apps/fortio-deploy created
PS C:\WINDOWS\system32> $env:FORTIO_POD = $(kubectl get pods -l app=fortio -o 'jsonpath={.items[0].metadata.name}')
PS C:\WINDOWS\system32> echo $env:FORTIO_POD
fortio-deploy-7647467df4-8jtgg
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl get destinationrule httpbin
NAME      HOST      AGE
httpbin   httpbin   54s
PS C:\WINDOWS\system32>
```

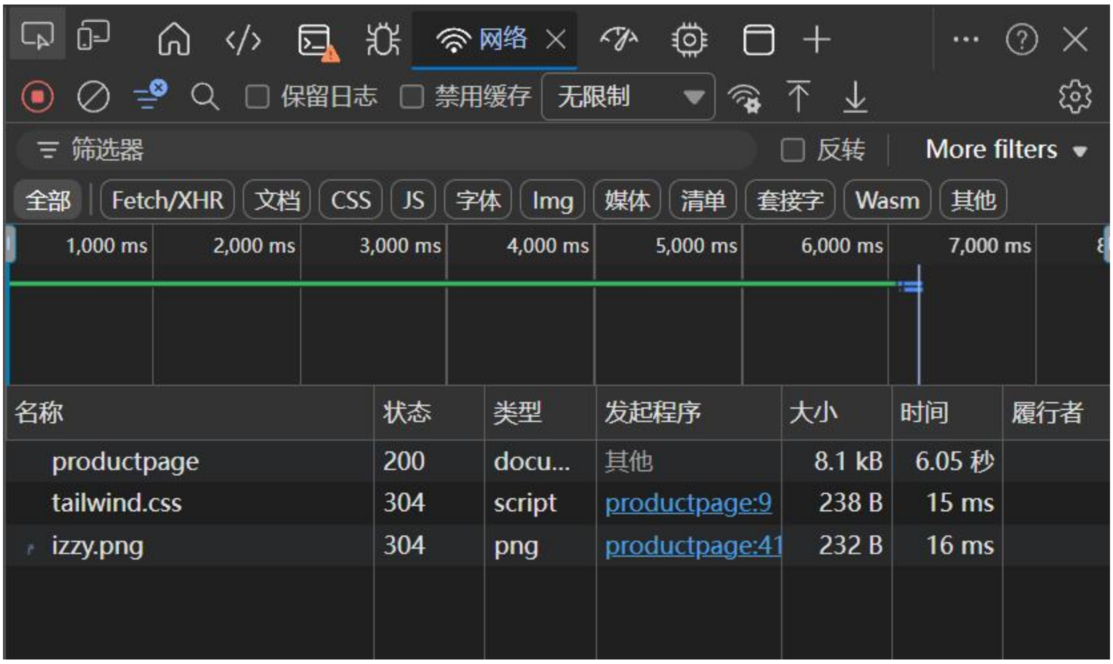
```
PS C:\WINDOWS\system32> @'
>> apiVersion: networking.istio.io/v1
>> kind: DestinationRule
>> metadata:
>>   name: httpbin
>> spec:
>>   host: httpbin
>>   trafficPolicy:
>>     connectionPool:
>>       tcp:
>>         maxConnections: 1
>>       http:
>>         http1MaxPendingRequests: 1
>>         maxRequestsPerConnection: 1
>>     outlierDetection:
>>       consecutive5xxErrors: 1
>>       interval: 1s
>>       baseEjectionTime: 3m
>>       maxEjectionPercent: 100
>> '@ | Out-File -FilePath "$env:USERPROFILE\httpbin-circuit-breaking.yaml" -Encoding utf8
PS C:\WINDOWS\system32> kubectl apply -f "$env:USERPROFILE\httpbin-circuit-breaking.yaml"
destinationrule.networking.istio.io/httpbin created
PS C:\WINDOWS\system32>
```

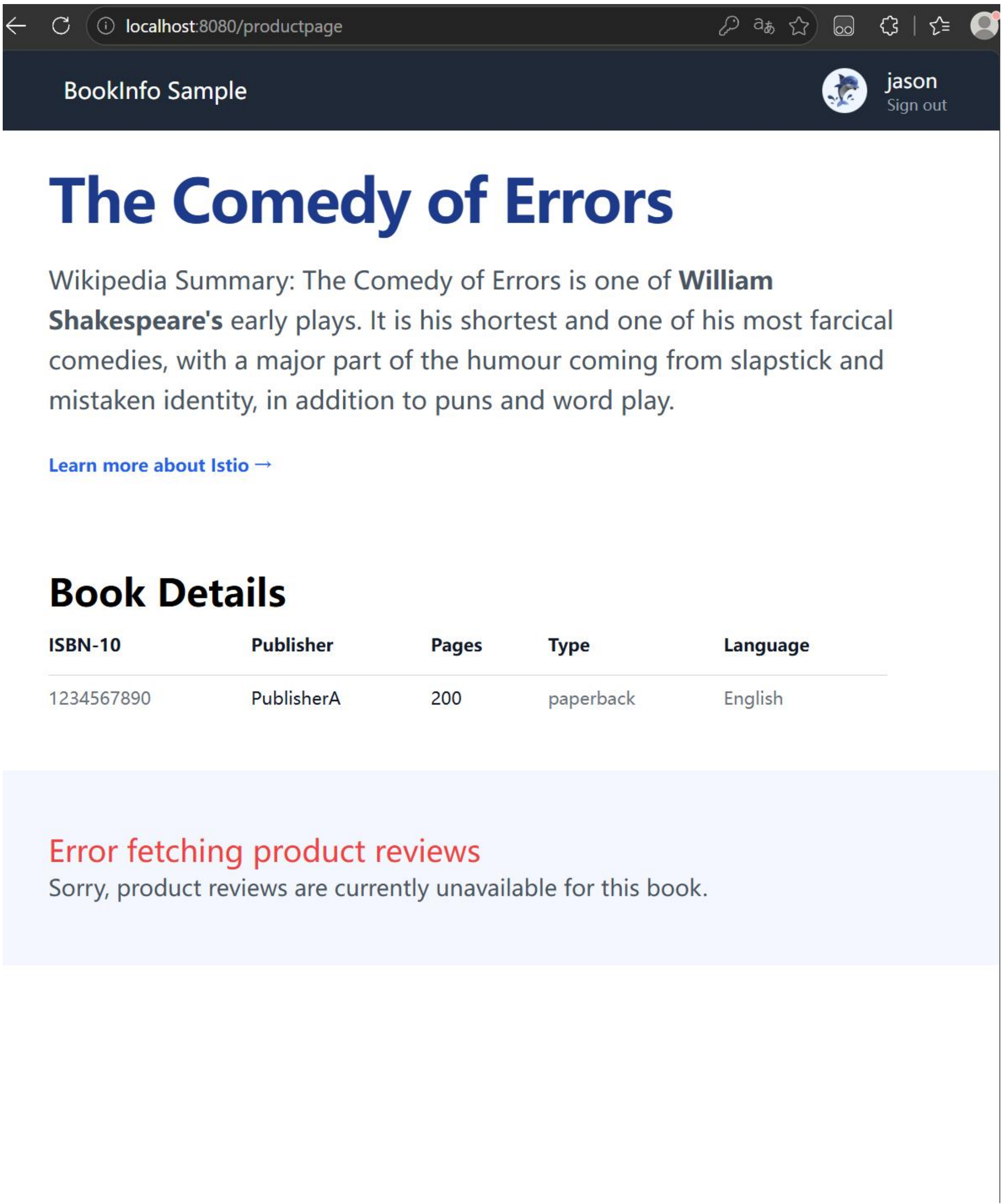
```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/httpbin/http
bin.yaml
serviceaccount/httpbin created
service/httpbin created
deployment.apps/httpbin created
PS C:\WINDOWS\system32> kubectl get pods -l app=httpbin
NAME                       READY   STATUS    RESTARTS   AGE
httpbin-654df84766-t75sx   2/2     Running   0          6s
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl edit virtualservice ratings
virtualservice.networking.istio.io/ratings edited
PS C:\WINDOWS\system32> kubectl get virtualservice ratings -o yaml | findstr fixedDelay

      {"apiVersion":"networking.istio.io/v1","kind":"VirtualService","metadata":{"annotations":{},"name":"ratings","name
space":"default"},"spec":{"hosts":["ratings"],"http":[{"fault":{"delay":{"fixedDelay":"7s","percentage":{"value":100}}},
"match":[{"headers":{"end-user":{"exact":"jason"}}}],"route":[{"destination":{"host":"ratings","subset":"v1"}}]},{"route
":[{"destination":{"host":"ratings","subset":"v1"}}]}]}}
        fixedDelay: 2s
PS C:\WINDOWS\system32> |
```

```
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
#
apiVersion: networking.istio.io/v1
kind: VirtualService
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |

{"apiVersion":"networking.istio.io/v1","kind":"VirtualService","metadata":{"annotations":{},"name":"ratings","namespace":"defaul
t"},"spec":{"hosts":["ratings"],"http":[{"fault":{"delay":{"fixedDelay":"7s","percentage":{"value":100}}},"match":[{"headers":{"end-
user":{"exact":"jason"}}}],"route":[{"destination":{"host":"ratings","subset":"v1"}}]},{"route":[{"destination":{"host":"ratings","subset"
:"v1"}}]}]}}
  creationTimestamp: "2026-05-23T08:07:09Z"
  generation: 2
  name: ratings
  namespace: default
  resourceVersion: "34916"
  uid: bba612c7-d4b0-4936-ad1e-6fb726747ff1
spec:
  hosts:
  - ratings
  http:
  - fault:
      delay:
        fixedDelay: 7s
        percentage:
          value: 100
    match:
    - headers:
        end-user:
          exact: jason
```

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/virtual-service-reviews-v3.yaml
virtualservice.networking.istio.io/reviews configured
PS C:\WINDOWS\system32> kubectl edit virtualservice ratings
PS C:\WINDOWS\system32> |
```


网络
保留日志
禁用缓存
无限制
筛选器
反转
More filters
全部
Fetch/XHR
文档
CSS
JS
字体
Img
媒体
清单
套接字
Wasm
其他
1,000 ms
2,000 ms
3,000 ms
4,000 ms
5,000 ms
6,000 ms
7,000 ms
名称
状态
类型
发起程序
大小
时间
履行者
productpage
200
docu...
其他
8.1 kB
6.05 秒
tailwind.css
304
script
productpage:9
238 B
15 ms
izzy.png
304
png
productpage:41
232 B
16 ms


localhost:8080/productpage
BookInfo Sample
jason
Sign out
The Comedy of Errors
Wikipedia Summary: The Comedy of Errors is one of William Shakespeare's early plays. It is his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.
Learn more about Istio →
Book Details
ISBN-10 Publisher Pages Type Language
1234567890 PublisherA 200 paperback English
Error fetching product reviews
Sorry, product reviews are currently unavailable for this book.

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/virtual-service-ratings-test-delay.yaml
virtualservice.networking.istio.io/ratings unchanged
PS C:\WINDOWS\system32> kubectl get virtualservice ratings -o yaml
apiVersion: networking.istio.io/v1
kind: VirtualService
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"networking.istio.io/v1","kind":"VirtualService","metadata":{"annotations":{},"name":"ratings","name
space":"default"},"spec":{"hosts":["ratings"],"http":[{"fault":{"delay":{"fixedDelay":"7s","percentage":{"value":100}}},
"match":[{"headers":{"end-user":{"exact":"jason"}}}],"route":[{"destination":{"host":"ratings","subset":"v1"}}]},{"route
":[{"destination":{"host":"ratings","subset":"v1"}}]}]}}
  creationTimestamp: "2026-05-23T08:07:09Z"
  generation: 2
  name: ratings
  namespace: default
  resourceVersion: "34916"
  uid: bba612c7-d4b0-4936-ad1e-6fb726747ff1
spec:
  hosts:
  - ratings
  http:
  - fault:
      delay:
        fixedDelay: 7s
        percentage:
          value: 100
    match:
    - headers:
        end-user:
          exact: jason
    route:
    - destination:
        host: ratings
        subset: v1
  - route:
    - destination:
        host: ratings
        subset: v1
```

```
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/httpbin/sam
ple-client/fortio-deploy.yaml
service "fortio" deleted from default namespace
deployment.apps "fortio-deploy" deleted from default namespace
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/httpbin/htt
pbin.yaml
serviceaccount "httpbin" deleted from default namespace
service "httpbin" deleted from default namespace
deployment.apps "httpbin" deleted from default namespace
PS C:\WINDOWS\system32> kubectl delete destinationrule httpbin
Error from server (NotFound): destinationrules.networking.istio.io "httpbin" not found
```

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/httpbin/samp
le-client/fortio-deploy.yaml
service/fortio created
deployment.apps/fortio-deploy created
```

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/httpbin/http
bin.yaml
serviceaccount/httpbin created
service/httpbin created
deployment.apps/httpbin created
```

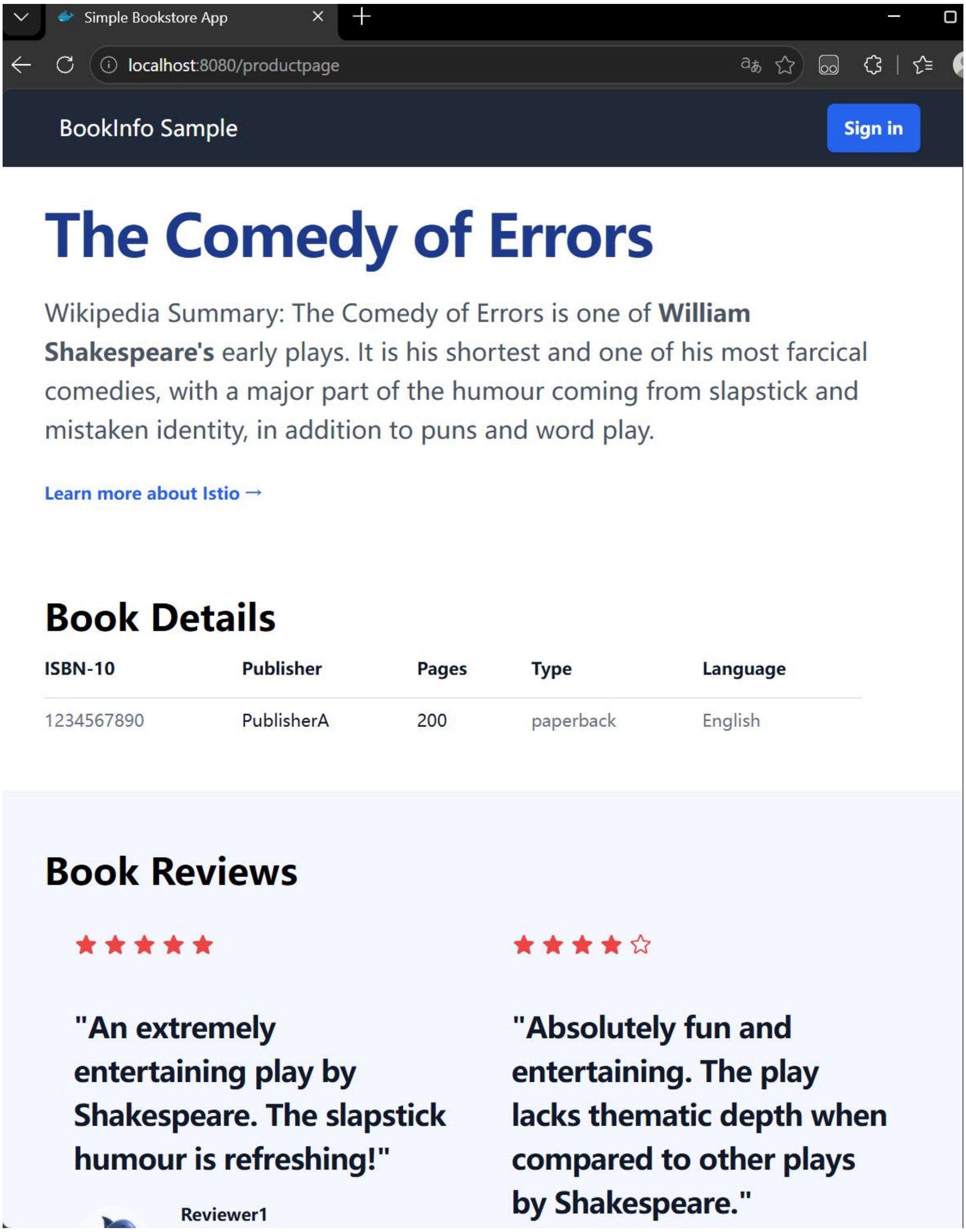
```
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/ne
tworking/virtual-service-ratings-test-abort.yaml
virtualservice.networking.istio.io "ratings" deleted from default namespace
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/ne
tworking/virtual-service-reviews-test-v2.yaml
virtualservice.networking.istio.io "reviews" deleted from default namespace
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/ne
tworking/virtual-service-all-v1.yaml
virtualservice.networking.istio.io "productpage" deleted from default namespace
virtualservice.networking.istio.io "details" deleted from default namespace
Error from server (NotFound): error when deleting "https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bo
okinfo/networking/virtual-service-all-v1.yaml": virtualservices.networking.istio.io "reviews" not found
Error from server (NotFound): error when deleting "https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bo
okinfo/networking/virtual-service-all-v1.yaml": virtualservices.networking.istio.io "ratings" not found
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/ne
tworking/virtual-service-ratings-test-delay.yaml
virtualservice.networking.istio.io "ratings" deleted from default namespace
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/virtual-service-ratings-test-abort.yaml
virtualservice.networking.istio.io/ratings created
PS C:\WINDOWS\system32>
```

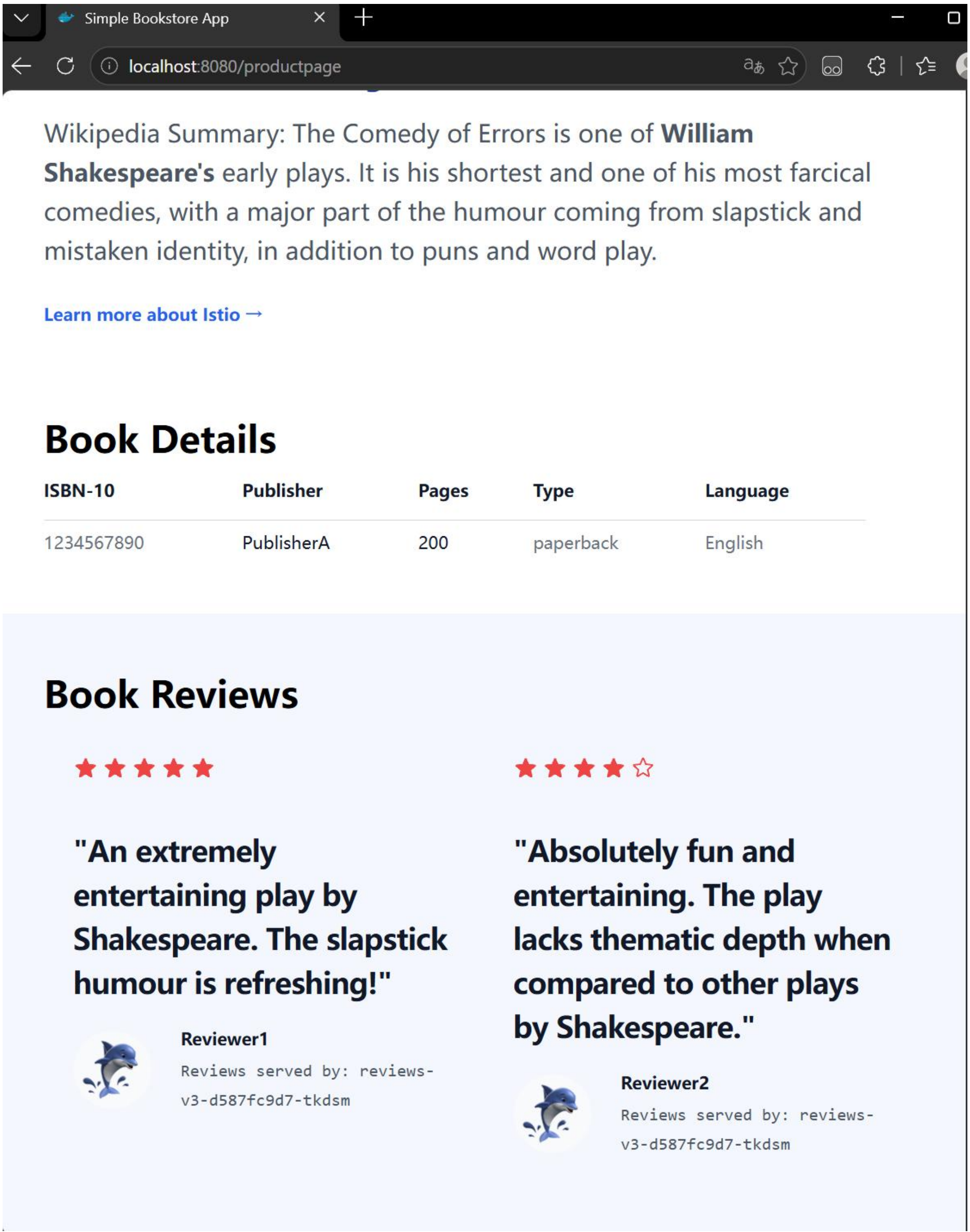
```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/virtual-service-ratings-test-delay.yaml
virtualservice.networking.istio.io/ratings configured
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/virtual-service-all-v1.yaml
virtualservice.networking.istio.io/productpage unchanged
virtualservice.networking.istio.io/reviews created
virtualservice.networking.istio.io/ratings unchanged
virtualservice.networking.istio.io/details unchanged
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/virtual-service-reviews-test-v2.yaml
virtualservice.networking.istio.io/reviews configured
PS C:\WINDOWS\system32>
```

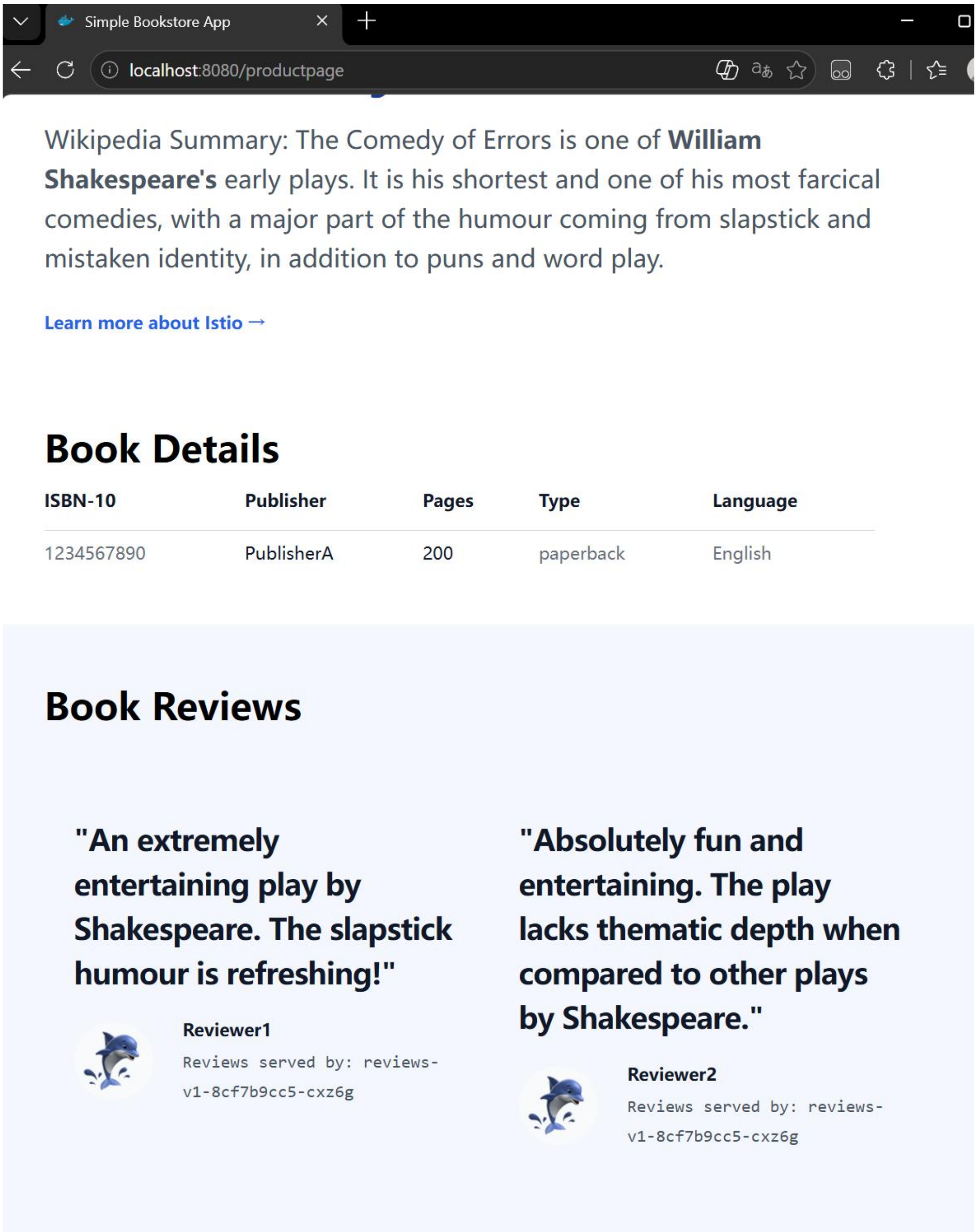
```
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/ne
tworking/virtual-service-reviews-v3.yaml
virtualservice.networking.istio.io "reviews" deleted from default namespace
PS C:\WINDOWS\system32>
```




```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/
release-1.29/samples/bookinfo/networking/virtual-service-reviews-v3.yaml
virtualservice.networking.istio.io/reviews configured
PS C:\WINDOWS\system32>
```


Simple Bookstore App
localhost:8080/productpage
Wikipedia Summary: The Comedy of Errors is one of William Shakespeare's early plays. It is his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.
Learn more about Istio →
Book Details
ISBN-10
Publisher
Pages
Type
Language
1234567890
PublisherA
200
paperback
English
Book Reviews
"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"
Reviewer1
Reviews served by: reviews-v3-d587fc9d7-tkdsm
"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."
Reviewer2
Reviews served by: reviews-v3-d587fc9d7-tkdsm

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/virtual-service-reviews-50-v3.yaml
virtualservice.networking.istio.io/reviews configured
PS C:\WINDOWS\system32> kubectl get virtualservice reviews -o yaml
apiVersion: networking.istio.io/v1
kind: VirtualService
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"networking.istio.io/v1","kind":"VirtualService","metadata":{"annotations":{},"name":"reviews","name
space":"default"},"spec":{"hosts":["reviews"],"http":[{"route":[{"destination":{"host":"reviews","subset":"v1"},"weight"
:50},{"destination":{"host":"reviews","subset":"v3"},"weight":50}]}]}}
  creationTimestamp: "2026-05-23T07:50:37Z"
  generation: 2
  name: reviews
  namespace: default
  resourceVersion: "32779"
  uid: 0d30a728-3b16-4e6b-b122-b0d51777101b
spec:
  hosts:
  - reviews
  http:
  - route:
    - destination:
        host: reviews
        subset: v1
      weight: 50
    - destination:
        host: reviews
        subset: v3
      weight: 50
PS C:\WINDOWS\system32> |
```

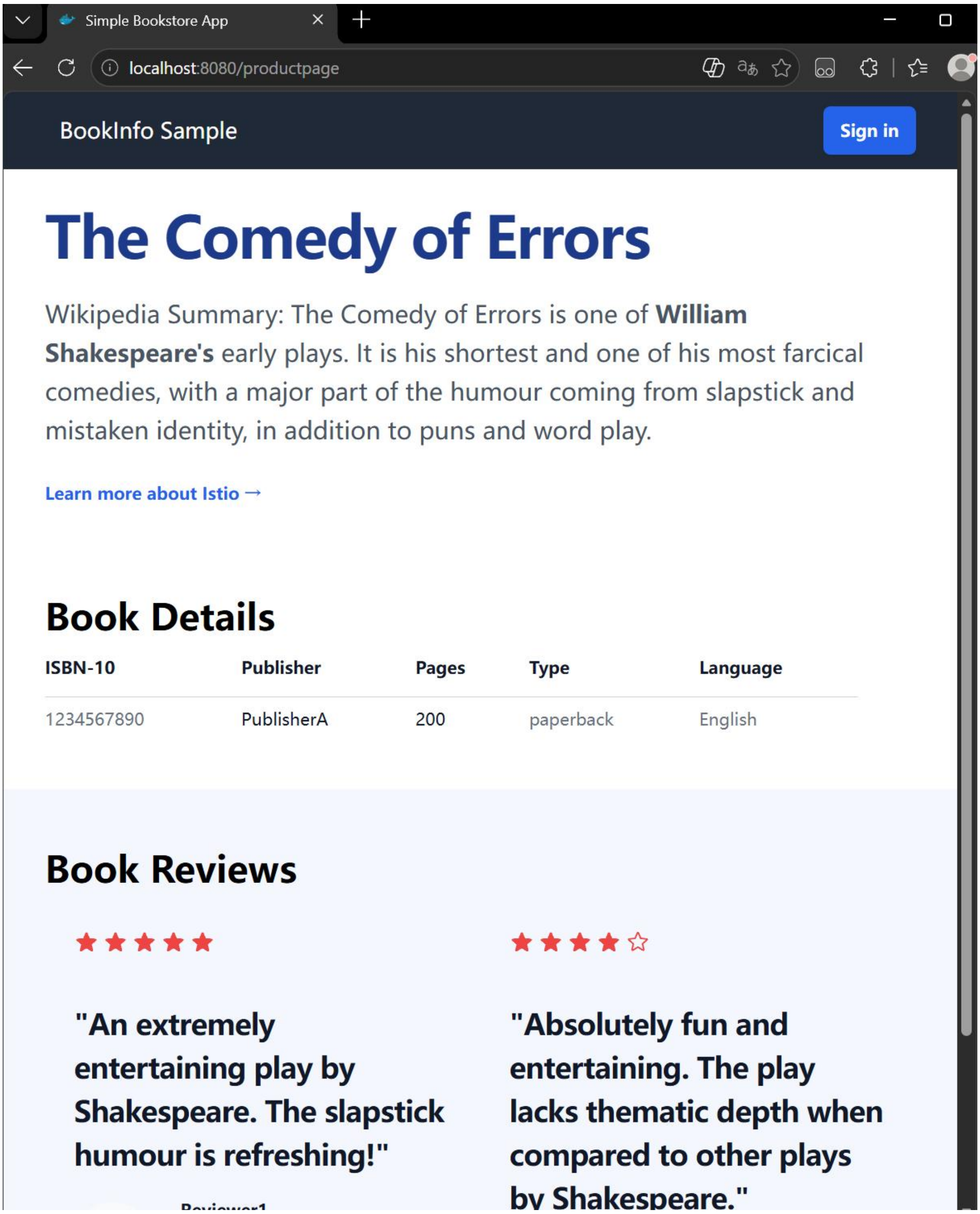



```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/networking/virtual-service-all-v1.yaml
virtualservice.networking.istio.io/productpage created
virtualservice.networking.istio.io/reviews created
virtualservice.networking.istio.io/ratings created
virtualservice.networking.istio.io/details created
PS C:\WINDOWS\system32> kubectl get virtualservices
NAME          GATEWAYS   HOSTS             AGE
details                  ["details"]       8s
productpage              ["productpage"]   8s
ratings                  ["ratings"]       8s
reviews                  ["reviews"]       8s
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/virtual-service-reviews-test-v2.yaml
virtualservice.networking.istio.io/reviews configured
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/ne
tworking/virtual-service-all-v1.yaml
virtualservice.networking.istio.io "productpage" deleted from default namespace
virtualservice.networking.istio.io "reviews" deleted from default namespace
virtualservice.networking.istio.io "ratings" deleted from default namespace
virtualservice.networking.istio.io "details" deleted from default namespace
PS C:\WINDOWS\system32> kubectl delete -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/ne
tworking/virtual-service-reviews-test-v2.yaml
Error from server (NotFound): error when deleting "https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bo
okinfo/networking/virtual-service-reviews-test-v2.yaml": virtualservices.networking.istio.io "reviews" not found
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/virtual-service-all-v1.yaml
virtualservice.networking.istio.io/productpage created
virtualservice.networking.istio.io/reviews created
virtualservice.networking.istio.io/ratings created
virtualservice.networking.istio.io/details created
PS C:\WINDOWS\system32> kubectl get virtualservices
NAME          GATEWAYS   HOSTS             AGE
details                  ["details"]       9s
productpage              ["productpage"]   9s
ratings                  ["ratings"]       9s
reviews                  ["reviews"]       9s
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl apply -f https://raw.githubusercontent.com/istio/istio/release-1.29/samples/bookinfo/net
working/destination-rule-all-mtls.yaml
destinationrule.networking.istio.io/productpage created
destinationrule.networking.istio.io/reviews created
destinationrule.networking.istio.io/ratings created
destinationrule.networking.istio.io/details created
PS C:\WINDOWS\system32> kubectl get destinationrule
NAME          HOST          AGE
details       details       9s
productpage   productpage   9s
ratings       ratings       9s
reviews       reviews       9s
PS C:\WINDOWS\system32>
```


Simple Bookstore App
localhost:8080/productpage
BookInfo Sample
Sign in
The Comedy of Errors
Wikipedia Summary: The Comedy of Errors is one of William Shakespeare's early plays. It is his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.
Learn more about Istio →
Book Details
ISBN-10
Publisher
Pages
Type
Language
1234567890
PublisherA
200
paperback
English
Book Reviews
"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"
"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."

```
Windows PowerShell
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安装最新的 PowerShell，了解新功能和改进！https://aka.ms/PSWindows

PS C:\WINDOWS\system32> kubectl get svc bookinfo-gateway-istio
NAME                      TYPE        CLUSTER-IP      EXTERNAL-IP    PORT(S)             AG
E
bookinfo-gateway-istio    ClusterIP   10.96.180.33    <none>         15021/TCP,80/TCP    13
d
PS C:\WINDOWS\system32>
```

```
Windows PowerShell

Windows PowerShell
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安装最新的 PowerShell，了解新功能和改进！https://aka.ms/PSWindows

PS C:\WINDOWS\system32> kubectl port-forward svc/bookinfo-gateway-istio 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
Handling connection for 8080
```

```
PS C:\WINDOWS\system32> kubectl get svc -n istio-system
NAME                           TYPE           CLUSTER-IP      EXTERNAL-IP    PORT(S)
                                      AGE
grafana                        ClusterIP      10.96.115.110   <none>         3000/TCP
                                      13d
istio-ingressgateway           LoadBalancer   10.96.85.90     <pending>      15021:30983/
TCP,80:30413/TCP,443:31606/TCP        4m47s
istiod                         ClusterIP      10.96.67.35     <none>         15010/TCP,15
012/TCP,443/TCP,15014/TCP             13d
istiod-revision-tag-default    ClusterIP      10.96.37.89     <none>         15010/TCP,15
012/TCP,443/TCP,15014/TCP             13d
jaeger-collector               ClusterIP      10.96.124.235   <none>         14268/TCP,14
250/TCP,9411/TCP,4317/TCP,4318/TCP    13d
kiali                          ClusterIP      10.96.95.181    <none>         20001/TCP,90
90/TCP                                13d
prometheus                     ClusterIP      10.96.127.234   <none>         9090/TCP
                                      13d
tracing                        ClusterIP      10.96.111.46    <none>         80/TCP,16685
/TCP                                  13d
zipkin                         ClusterIP      10.96.141.153   <none>         9411/TCP
                                      13d
PS C:\WINDOWS\system32> kubectl -n istio-system get service istio-ingressgateway -o jso
npath='{.spec.ports[?(@.name==\"http2\")].port}'
80
PS C:\WINDOWS\system32> kubectl get gateway
NAME               CLASS    ADDRESS                                             PROGRAMME
D   AGE
bookinfo-gateway   istio    bookinfo-gateway-istio.default.svc.cluster.local    True
    13d
PS C:\WINDOWS\system32> kubectl get virtualservice
No resources found in default namespace.
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> kubectl get svc istio-ingressgateway -n istio-system
NAME                   TYPE           CLUSTER-IP    EXTERNAL-IP   PORT(S)
                        AGE
istio-ingressgateway   LoadBalancer   10.96.85.90   <pending>     15021:30983/TCP,80:30
413/TCP,443:31606/TCP   95s
PS C:\WINDOWS\system32>
```